

Highlights

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The generalized Lie derivative for the natural stress tensor[★]

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ABSTRACT

Numerical simulations of viscoelastic flows is classically restricted by the High Weissenberg Number Problem (HWNP), primarily driven by advective instabilities within the Upper-Convected Time Derivative (UCTD) of standard approaches. To overcome this limitation, here we introduce a method that combines the Generalized Lie Derivative and the Natural Stress Tensor Formulation (GLD-NSF). By expressing the Constitutive Equation in natural coordinates and applying the generalized Lie derivative in the UCTD expression, the proposed method intends to circumvent the numerical instabilities arising with the calculation of advective component and the localized singularity growths that typically affect standard methods. The robustness and accuracy of the GLD-NSF are rigorously evaluated considering benchmark problems available in literature using the Oldroyd-B. Numerical results for the lid-driven cavity are also addressed to demonstrate that while the conventional methods fails to converge at $Wi = 1.0$, the GLD-NSF maintains numerical stability, yielding mesh-independent stress fields without spurious oscillations. Furthermore, the global flow topology and localized stress profiles exhibit excellent quantitative agreement with established stabilization techniques in the literature, such as the Log-Conformation and Square-Root-Conformation methods. Ultimately, the GLD-NSF provides an elegant, mathematically rigorous alternative to circumvent the HWNP without relying on complex stabilization schemes.

1. Introduction

Most viscoelastic fluid models incorporate the Upper-Convected Time Derivative (UCTD) Bird, Armstrong and Hassager (1986) to ensure material frame invariance—the principle that physical laws remain consistent regardless of the frame of reference of the observer. For fluids characterized by memory, the standard material derivative is insufficient as it fails to account for the local deformation and rotation of the polymer microstructure. The UCTD of the stress tensor $\mathbf{T}$, denoted as $\overset{\nabla}{\mathbf{T}}$, includes a non-linear advection term $(\mathbf{u} \cdot \nabla)\mathbf{T}$ that imparts a first-order hyperbolic character to the constitutive equations. Consequently, stress information propagates as waves along streamlines rather than diffusing isotropically, which often leads to numerical instabilities such as shocks or sharp discontinuities within the stress field.

To mitigate these hyperbolic instabilities, researchers typically employ specialized upwind schemes such as First Order Upwind (FOU) Spalding (1972) or high-resolution schemes like ADBQUICKEST Ferreira, Kurokawa, Queiroz, Kaibara, Oishi, Cuminato, Castelo, Tomé and McKee (2009)

and CUBISTA Alves, Oliveira and Pinho (2003). Furthermore, the advective nature of the UCTD imposes a Courant-Friedrichs-Lewy (CFL) condition De Moura and Kubrusly (2013), which acts as a strict limitation on the allowable time step. A more profound challenge is the High Weissenberg Number Problem (HWNP) Keunings (1986), characterized by the exponential growth of stress gradients near boundaries and stagnation points. Standard numerical methods based on polynomials struggle to resolve these steep gradients, often resulting in a total loss of convergence. A landmark solution to this difficulty is the Log-Conformation Reformulation (LCR) Fattal and Kupferman (2004, 2005), which solves for the logarithm of the conformation tensor $\mathbf{A}$, transforming the exponential growth of the stress into a more manageable linear growth. In addition to addressing the exponential growth of stress, the LCR preserves the positive-definiteness of the conformation tensor $\mathbf{A}$. The loss of this physical property is often identified as a primary cause of the instabilities associated with the HWNP.

Recent advancements have introduced hybrid Eulerian-Lagrangian approaches, such as the Generalized Lie Derivative (GLD) Lee and Xu (2006); Medeiros, Notsu and Oishi (2021). The GLD replaces the UCTD with a temporal derivative in a Lagrangian framework while maintaining an underlying Eulerian mesh. By tracking the trajectories of the particles and interpolating properties, this strategy has been shown to eliminate the CFL restriction and alleviate the effects of the HWNP Medeiros et al. (2021). Parallel to this approach is the Natural Stress Formulation (NSF) Renardy (1994); Afonso, Evans and Palhares Junior (2025), which aligns the components of the stress with the local

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streamlines. Originally designed to stabilize flow around corners, the NSF has demonstrated significant potential in capturing asymptotic behavior near singularities and improving convergence at high Weissenberg numbers Evans, Palhares Junior, Oishi and Ruano Neto (2022).

In this work, a novel numerical framework is proposed that integrates these two powerful strategies: the GLD-NSF formulation. By substituting the UCTD with the GLD and simultaneously aligning the stress tensor with the streamlines via the NSF, this method leverages the stability of Lagrangian tracking alongside the geometric advantages of the natural stress basis. The primary objective of this study is to present the mathematical derivation of the GLD-NSF formulation and validate its performance using a manufactured flow solution and the benchmark case of the lid-driven cavity flow.

2. Governing Equations

The governing equations are composed by the mass and momentum conservation equations with a constitutive equation that describes the behaviour of the viscoelastic fluid of interest, in this case the Oldroyd-B flow, assumed as incompressible. The equations are made dimensionless using the characteristic dimensional parameters of length L_c , velocity U_c , and the total zero-shear viscosity $\eta_0 = \eta_s + \eta$, where η_s and η are the solvent and the polymeric components of viscosity, respectively. The resulting set of dimensionless equations is given by

$$\begin{cases} \nabla \cdot \mathbf{u} = 0 \\ \frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \left(\frac{\beta}{Re}\right) \nabla^2 \mathbf{u} + \left(\frac{1}{Re}\right) \nabla \cdot \mathbf{T} \\ \mathbf{T} + \overset{\nabla}{Wi} \mathbf{T} = 2(1 - \beta) \mathbf{D} \end{cases} \quad (1)$$

where $\mathbf{u}$ represents the velocity vector field, p is the pressure field. The term $\mathbf{D}$ is the rate-of-deformation tensor, $\mathbf{D} = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T)$. In equations, $\mathbf{T}$ represents the polymeric component of the total extra stress tensor, $\boldsymbol{\sigma}$ ($\boldsymbol{\sigma} = \mathbf{T}_s + \mathbf{T}$, where $\mathbf{T}_s$ represents the Newtonian solvent component). The superscript ' ∇ ' denotes the Upper-Convected Time Derivative operator (UCTD) Oldroyd (1950) that is addressed in the next section.

The non-dimensional parameters in equations are the Reynolds number (Re), the Weissenberg number (Wi) and the viscosity ratio (β) defined by

$$Re = \frac{\rho U_c L_c}{\eta_0}, \quad Wi = \frac{\lambda U_c}{L_c}, \quad \beta = \frac{\eta_s}{\eta_0}, \quad (2)$$

where λ is the fluid relaxation time.

2.1. The Upper Convected Time Derivative (UCTD)

As mentioned in the introduction, handling of the UCTD is critical for numerical stability, particularly at High Weissenberg Number Problems (HWNPs). In this section, we

detail the progression from standard Eulerian techniques to the proposed Lagrangian, stream-aligned hybrid scheme we developed.

2.1.1. Eulerian Framework: Cartesian Stress Formulation (CSF)

The traditional approach, referred to as the Cartesian Stress Formulation (CSF), expands the UCTD in a fixed Eulerian coordinate system. The derivative takes the expanded form Joseph (1990):

$$\overset{\nabla}{\mathbf{T}} := \frac{\partial \mathbf{T}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{T} - \nabla \mathbf{u} \mathbf{T} - \mathbf{T} \nabla \mathbf{u}^T. \quad (3)$$

While computationally straightforward, directly discretizing this expanded form can trigger severe numerical instabilities at high Weissenberg numbers. This numerical instabilities or the break down of the simulation is caused by a failure of the method to balance the exponential growth of the stress due the deformation with the convective term, which is also related to the loss of the positivity definite property of the conformation tensor $\mathbf{A}$ Afonso, Pinho and Alves (2012).

2.1.2. Lagrangian Framework: Generalized Lie Derivative (GLD)

To circumvent the numerical stability limitations associated with solving the UCTD in an Eulerian framework, we adopt a Lagrangian perspective where fluid elements are tracked along their trajectories. In this framework, the UCTD is reformulated through the Generalized Lie Derivative. We define the operator $\mathcal{L}$ such that $\overset{\nabla}{\mathbf{T}} \equiv \mathcal{L}\{\mathbf{T}\}$ Medeiros et al. (2021):

$$\mathcal{L}\{\mathbf{T}\} := \mathbf{L}_t(\tau) \frac{\partial}{\partial \tau} [\mathbf{L}_\tau(t) \mathbf{T}(\tau) \mathbf{L}_\tau(t)^T] \mathbf{L}_t(\tau)^T, \quad (4)$$

where the transition matrices are defined as

$$\mathbf{L}_t(\tau) := \nabla \chi_t(\mathbf{x}, \tau) \quad \text{and} \quad \mathbf{L}_\tau(t) := \nabla \chi_\tau(\chi, t). \quad (5)$$

Here, $\chi_t(\mathbf{x}, \tau)$ represents the mapping of a fluid element position χ at time τ relative to a reference position $\mathbf{x}$ and time t , such that $\chi = \chi_t(\mathbf{x}, \tau)$ and, conversely, $\mathbf{x} = \chi_\tau(\chi, t)$. We evaluate the fluid element deformation through the transition matrix $\mathbf{L}_t(\tau) := \partial \chi / \partial \mathbf{x}$ ($[\mathbf{L}_t(\tau)]_{ij} = \partial \chi_i / \partial x_j$). Differentiating with respect to τ and applying the chain rule, we obtain:

$$\frac{\partial \mathbf{L}_t(\tau)}{\partial \tau} = \frac{\partial \mathbf{u}(\tau)}{\partial \mathbf{x}} = \frac{\partial \mathbf{u}(\tau)}{\partial \chi} \frac{\partial \chi}{\partial \mathbf{x}}, \quad (6)$$

which simplifies to

$$\frac{\partial \mathbf{L}_t(\tau)}{\partial \tau} = \nabla \mathbf{u}(\tau) \mathbf{L}_t(\tau). \quad (7)$$

Regarding the inverse mapping, the identity $\mathbf{L}_t(\tau)\mathbf{L}_t(t) = \mathbf{I}$ must hold. By differentiating this relation with respect to τ and substituting Eq. (7), we determine the temporal evolution of the inverse transition matrix $\partial\mathbf{L}_t(t)/\partial\tau = -\mathbf{L}_t(t)\nabla\mathbf{u}(\tau)$.

The transport of the stress tensor, in Lagrangian frame, is obtained by a pull-back/push-forward operation. The deformation of the fluid element is measured along its trajectory and accounted in the transition matrix. This avoids the calculation of the stress tensor transport directly by the convective terms as in CSF.

2.1.3. The Natural Stress Formulation (NSF)

Another strategy that could circumvent the numerical instabilities of HWNP is by adopting a coordinate system that is not fixed as the standard Cartesian one but more likely aligned to the principal components of the stress. This is achieved by rewriting the stress tensor $\mathbf{T}$ to consider a stream-aligned coordinate system that changes as the fluid flows. The resulting method is referred here as the Natural Stress Formulation (NSF).

For the NSF, we define a rotation matrix, $\mathbf{R}$, used to transform the standard Cartesian basis vectors into a natural, stream-aligned basis. It also convenient to define the spin tensor, $\mathcal{R}$, that indicates how $\mathbf{R}$ changes along the flow. In two dimensions, the Rotation and the Spin tensors are given as

$$\mathbf{R} := \frac{1}{|\mathbf{u}|} \begin{pmatrix} u & -v \\ v & u \end{pmatrix}, \quad \mathcal{R} := \mathbf{R}^\top \frac{D\mathbf{R}}{Dt}, \quad (8)$$

where u and v are the Cartesian velocity components ($\mathbf{u} = u\hat{\mathbf{i}} + v\hat{\mathbf{j}} + 0\hat{\mathbf{k}}$), the velocity magnitude is calculated as $|\mathbf{u}| = \sqrt{u^2 + v^2}$ and D/Dt indicates the material derivative. As easily noticed, the rotation matrix is a proper orthogonal matrix and then, it holds the properties $\mathbf{R}^\top \mathbf{R} = \mathbf{R}\mathbf{R}^\top = \mathbf{I}$ and $\det(\mathbf{R}) = 1$. From the material derivative of $\mathbf{R}^\top \mathbf{R} = \mathbf{I}$, it follows that $\mathcal{R}$ is a skew-symmetric tensor ($\mathcal{R}^\top = -\mathcal{R}$) Afonso et al. (2012).

The stream-aligned representation of the UCTD using the Natural Stress Formulation (NSF) is achieved by initially expressing the Cartesian UCTD in terms of the conformation tensor, $\mathbf{A}$, defined as $\mathbf{A} := \frac{Wi}{(1-\beta)}\mathbf{T} + \mathbf{I}$. This yields

$$\overset{\nabla}{\mathbf{T}} = \frac{(1-\beta)}{Wi} \left(\frac{D\mathbf{A}}{Dt} - \nabla\mathbf{u}\mathbf{A} - \mathbf{A}\nabla\mathbf{u}^\top + 2\mathbf{D} \right). \quad (9)$$

The transformation to the natural basis is made by defining the Natural Stress Tensor, $\mathbf{N} := \mathbf{R}^\top \mathbf{A} \mathbf{R}$, where the local rotation of the coordinate system is accounted using the rotation matrix $\mathbf{R}$.

By taking the material derivative of $\mathbf{A} = \mathbf{R}\mathbf{N}\mathbf{R}^\top$ and incorporating the spin tensor $\mathcal{R}$, we effectively decouple the rotational kinematics from the deformation. If we further defining the transformed velocity gradient in the natural basis as $\widetilde{\nabla}\mathbf{u} := \mathbf{R}^\top \nabla\mathbf{u} \mathbf{R} - \mathcal{R}$ (which transposed becomes $\widetilde{\nabla}\mathbf{u}^\top = \mathbf{R}^\top \nabla\mathbf{u}^\top \mathbf{R} + \mathcal{R}$), we finally get the UCTD is given by

$$\overset{\nabla}{\mathbf{T}} = \frac{(1-\beta)}{Wi} \mathbf{R} \left(\frac{D\mathbf{N}}{Dt} - \widetilde{\nabla}\mathbf{u} \mathbf{N} - \mathbf{N} \widetilde{\nabla}\mathbf{u}^\top + 2\widetilde{\mathbf{D}} \right) \mathbf{R}^\top, \quad (10)$$

where the transformed rate-of-deformation is given by $\widetilde{\mathbf{D}} = (\widetilde{\nabla}\mathbf{u} + \widetilde{\nabla}\mathbf{u}^\top)/2$, or equivalently, $\mathbf{D} = \mathbf{R}\widetilde{\mathbf{D}}\mathbf{R}^\top$.

2.1.4. The Proposed GLD-NSF Method

The primary focus of the current paper is to develop a hybrid method combining the Lagrangian path-history tracking from GLD, with the streamline-aligned rotation of the coordinate basis from NSF. For that we define a transformed or natural transition matrix

$$\widetilde{\mathbf{L}}_t(\tau) := \mathbf{R}(\tau)^\top \mathbf{L}_t(\tau) \quad (11)$$

By taking the time derivative and applying the properties of the rotation and spin tensors, we find an expression similar to that in equation (7) but in natural coordinates

$$\frac{\partial \widetilde{\mathbf{L}}_t(\tau)}{\partial \tau} = \widetilde{\nabla}\mathbf{u}(\tau) \widetilde{\mathbf{L}}_t(\tau). \quad (12)$$

This system allows us to rewrite the UCTD completely in terms of the Natural Stress Tensor $\mathbf{N}$.

3. Numerical Methods

3.1. Discretized Constitutive Equations

The time-discretized constitutive equation for the viscoelastic stress considering each of the upper-convected time derivative formulation presented previously are then given by:

- CSF:

$$\mathbf{T}^{n+1} = \mathbf{T}^n + \Delta t \left(\nabla\mathbf{u} \mathbf{T}^n + \mathbf{T}^n \nabla\mathbf{u}^\top - \frac{1}{Wi} \mathbf{T}^n \right) \quad (13)$$

- GLD:

$$\mathbf{T}^{n+1} = \frac{Wi}{\Delta t + Wi} (\mathbf{L} \mathbf{T}^n \mathbf{L}^\top) + \frac{\Delta t}{\Delta t + Wi} \mathbf{I} \quad (14)$$

- NSF:

$$\mathbf{N}^{n+1} = \mathbf{N}^n + \Delta t \left(\widetilde{\nabla}\mathbf{u} \mathbf{N}^n + \mathbf{N}^n \widetilde{\nabla}\mathbf{u}^\top - \frac{1}{Wi} (\mathbf{N}^n - \mathbf{I}) \right) \quad (15)$$

- GLD-NSF:

$$\mathbf{N}^{n+1} = \frac{Wi}{\Delta t + Wi} (\widetilde{\mathbf{L}} \mathbf{N}^n \widetilde{\mathbf{L}}^\top) + \frac{\Delta t}{\Delta t + Wi} \mathbf{I} \quad (16)$$

The transition matrix ($\mathbf{L}$ in GLD or $\widetilde{\mathbf{L}}$ in GLD-NSF) is obtained by numerically integrating the expression given in equation (7) or, equivalently, (12). Using the explicit backward Euler method, the transition matrix is obtained by $\mathbf{L} = \mathbf{I} + \Delta t \nabla\mathbf{u}$. For the present GLD-NSF method, however, we employed a semi-implicit method:

$$\widetilde{\mathbf{L}} = \left(\mathbf{I} - \frac{\Delta t}{2} \widetilde{\nabla}\mathbf{u} \right)^{-1} \left(\mathbf{I} + \frac{\Delta t}{2} \widetilde{\nabla}\mathbf{u} \right). \quad (17)$$

4. Validation Tests

4.1. Manufactured problem

In this section we aim to test the GLD-NSF method with a manufactured bidimensional problem proposed by Venkatesan and Ganesan (2017). The velocity field is given by

$$\begin{aligned} u(x, y, t) &= \exp(-0.1t) \sin(\pi x), \\ v(x, y, t) &= -\pi \exp(-0.1t) y \cos(\pi x). \end{aligned}$$

For an Oldroyd-B fluid, the exact solution of the stress tensor is obtained analytically and given by the expressions:

$$\begin{aligned} T_{xx}(x, y, t) &= \exp(-0.1t) \sin(\pi x), \\ T_{xy}(x, y, t) &= -\pi \exp(-0.1t) y \cos(\pi x), \\ T_{yy}(x, y, t) &= \exp(-0.1t) \sin(\pi x) \cos(\pi y). \end{aligned}$$

The problem is defined in a domain $\Omega = [0, 1]^2$, with the dimensionless parameters $Wi = 0.25$ and $\beta = 0.75$. For this test, we employed an uniform mesh with $\Delta h = 0.05$ and set the time increment of $\Delta t = 0.005$. Figure 1 compares the results obtained by the GLD-NSF method with the exact solution of the stress tensor at time $t = 0.5$. Contours of the absolute errors at the same time are shown in Figure 2. The agreement with the exact solution shows the consistency of the GLD-NSF formulation.

4.2. Start-up rotating cylinder

Exact analytical solutions provide robust benchmarks for numerical validation. In this sub-section, we consider the transient start-up flow of an incompressible Oldroyd-B fluid confined within an infinitely long circular cylinder of radius R , as originally addressed by Fetecau (2005). At time $t = 0^+$, the cylinder starts rotating about its axis with a constant angular velocity, Ω . Consequently, the fluid, initially at rest, gradually moves in response to the shear induced by the rotating walls. The flow is axisymmetric, and the velocity field is strictly unidirectional, taking the form $\mathbf{v} = v(r, t)\mathbf{e}_\theta$.

To solve the problem, Fetecau (2005) decompose the velocity field into a steady-state component, which reduces to a rigid body rotation ($v(r, t \rightarrow \infty) = r$), and a transient component, $v_t(r, t)$. This transformation circumvents the non-homogeneous boundary conditions at the moving wall, yielding $v_t(1, t) = 0$ alongside the initial condition $v_t(r, 0) = -r$. Furthermore, applying the operator $(1 + Wi \frac{\partial}{\partial t})$ to the momentum equation and substituting the constitutive relation dictates the governing partial differential equation for $v_t(r, t)$. The exact solution for the full velocity field, $v(r, t) = r + v_t(r, t)$, then becomes

$$\begin{aligned} v(r, t) &= r + 2 \sum_{n=1}^{\infty} G_n(t) \frac{J_1(r_n r)}{r_n J_0(r_n)}, \\ \text{with } G_n(t) &= \frac{s_{n_2} \exp(s_{n_1} t) - s_{n_1} \exp(s_{n_2} t)}{(s_{n_2} - s_{n_1})}, \end{aligned} \quad (18)$$

where r_n are the positive roots of $J_1(r) = 0$, s_{n_1} and s_{n_2} are roots of the characteristic equation $(Re Wi)s_n^2 + (Re + \beta Wi r_n^2)s_n + r_n^2 = 0$.

Substituting the velocity solution and further integrating the constitutive equation along the tangential direction, we obtain the exact solution for $T_{r\theta}(r, t)$ given by

$$T_{r\theta}(r, t) = -2(1 - \beta) \sum_{n=1}^{\infty} K_n(t) \frac{J_2(r_n r)}{J_0(r_n)}, \quad (19)$$

$$\begin{aligned} \text{with } K_n(t) &= \frac{s_{n_2} [\exp(s_{n_1} t) - \exp(-t/Wi)]}{(1 + Wi s_{n_1})(s_{n_2} - s_{n_1})} \\ &- \frac{s_{n_1} [\exp(s_{n_2} t) - \exp(-t/Wi)]}{(1 + Wi s_{n_2})(s_{n_2} - s_{n_1})}. \end{aligned} \quad (20)$$

To comprehensively validate our numerical framework we also need the solution of the normal stress component, $T_{\theta\theta}(r, t)$. Since Fetecau (2005) did not derived an expression for it, we then extended the previous procedure to obtain the analytical solution of $T_{\theta\theta}(r, t)$. This resumes to evaluating the interaction between every pair of spatial modes governing the shear rate and the tangential stress.

$$T_{\theta\theta}(r, t) = 8(1 - \beta) \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \left[\frac{J_2(r_n r)}{J_0(r_n)} \frac{J_2(r_m r)}{J_0(r_m)} \right] H_{nm}(t), \quad (21)$$

$$H_{nm}(t) = \exp(-t/Wi) \int_0^t \exp(\tau/Wi) K_n(\tau) G_m(\tau) d\tau.$$

Since $K_n(\tau)$ and $G_m(\tau)$ are sums of exponentials, $H_{nm}(t)$ can be solved analytically by evaluating every combination of exponentials resulting from the product $K_n(\tau)G_m(\tau)$ and then using the rule

$$\begin{aligned} \mathcal{H}(A) &= \exp(-t/Wi) \int_0^t \exp(\tau/Wi) \exp(A\tau) d\tau \\ &= \frac{\exp(At) - \exp(-t/Wi)}{1 + AWi}. \end{aligned} \quad (22)$$

From these analytical results, we validated the GLD-NSF formulation. Figure 3 shows the temporal evolution of the velocity field alongside a direct comparison between the exact and numerical (GLD-NSF) solutions for the polymeric stresses $T_{r\theta}$ and $T_{\theta\theta}$. For these numerical tests, we set the Reynolds number to $Re = 100$, the Weissenberg number to $Wi = 0.5$, and the viscosity ratio to $\beta = 0.5$.

The method precision is remarkable. As corroborated by the area-weighted L^1 and L^2 error norms, which appropriately account for the axisymmetric geometry of the rotating cylinder, the GLD-NSF formulation demonstrates substantial agreement with the exact analytical series. Furthermore, our results substantiated that the GLD-NSF method circumvents the traditional instabilities observed in Eulerian frameworks, accurately capturing the transient stress dynamics of the Oldroyd-B fluid.

5. Numerical Tests

A numerical solver was also used to test the GLD-NSF formulation in more complex scenarios where analytical solutions are not available. For that, an in-house code written in C language was employed. A brief description of the code is given below but more details of the present code used in different cases can also be found in previous publications of the current group.

5.1. Numerical solver description

The code solves the incompressible Navier-Stokes equations on a staggered Marker-And-Cell (MAC) grid. The velocity components are defined at the cell edges while the pressure and stress at the cell center. The pressure and velocity fields are decoupled by using a fractional-step projection method. The temporal integration encompasses a predictor-corrector sequence. First, the intermediate velocity field, $\mathbf{u}^*$, is computed by solving the momentum equation without the divergence-free constraint. The viscous term is treated implicitly and the remaining terms explicitly.

The implicit Laplacian is discretized using second-order central finite differences. The explicit convective term is solved via the CUBISTA scheme, a third-order upwind-biased method that substantially mitigates hyperbolic instabilities. The resulting sparse linear system is solved by using the Krylov Subspace (KSP) solvers provided by the PETSc library.

Then, from the Helmholtz-Hodge decomposition, a Pressure-Poisson equation is used to find the scalar projection field, Ψ , ensuring the mass conservation constraint is satisfied $\nabla^2 \Psi = \nabla \cdot \mathbf{u}^*$. The intermediate velocity is then corrected to yield the final, divergence-free velocity field, and simultaneously update the pressure $\mathbf{u}^{n+1} = \mathbf{u}^* - \nabla \Psi$, $p^{n+1} = p^n + \frac{\Psi}{\Delta t}$. The stress tensor, $\mathbf{T}^{n+1}$, is then updated from the constitutive equation.

5.2. Lid-driven cavity

Here, we present results for the lid-driven cavity flow as done in the reference study of Venkatesan and Ganesan (2017). The domain consists in a two-dimensional unit square cavity, $\Omega = [0, 1]^2$, bounded by solid walls. Following the reference, we considered the regularized cavity case in which the horizontal motion of the top lid is such that the velocity gradient vanishes at the upper corners to avoid the non-physical stress singularities. The no-slip and no-penetration conditions are imposed along the boundaries and the horizontal velocity along the top lid is given by $u(x, t) = 8[1 + \tanh(8(t - 0.5))]x^2(1 - x)^2$ and $v = 0$. The fluid is assumed to be at rest in an isotropic state so the velocity field as the stress tensor are set to zero.

In order to reproduce one of the benchmark cases Venkatesan and Ganesan (2017), we set the very small Reynolds number of $Re = 0.01$, the Weissenberg number of $Wi = 0.5$ and the solvent viscosity ratio of $\beta = 0.5$. For the present results, the cases were simulated up to $t = 20.0$ with a time resolution of $\Delta t = 1 \times 10^{-4}$. The mesh is identical in both directions with the minimum spacing of

$\Delta h = 5 \times 10^{-4}$ at the boundaries and smoothly increased to $\Delta h = 5 \times 10^{-3}$ along a region of 20% of the cavity length. For the current comparison case, these parameters are such that produce results that are independent of the mesh refinement and temporal discretization.

Figure 5 shows the evolution of the elastic energy, $E_k = \frac{1}{2} \int_{\Omega} |\mathbf{u}(\mathbf{x}, t)|^2 dx$, for the CSF and GLD-NSF formulations with those presented in Venkatesan and Ganesan (2017). This plot shows the transient E_k solution and its following plateau of the steady flow obtained by the current simulations are virtually indistinguishable from the reference literature indicating that the overall behaviour of the flow dynamics is well reproduced by the code.

Figure 6 presents the streamlines obtained by CSF and GLD-NSF methods. The global flow topology encompasses a large primary vortex spanning the upper half of the cavity alongside two secondary, counter-rotating recirculating regions near the bottom corners. A direct comparison reveals that the streamlines generated by the GLD-NSF formulation (dashed lines) overlap perfectly with the standard CSF results (solid lines).

The markers in figure 6 indicate the location of the recirculating flow center, square for the CSF and cross for the GLD-NSF. Table 1 compares the location of the primary recirculation center of our simulations with those presented in literature. The GLD-NSF method predicts the recirculation eye at $(x_p, y_p) = (0.467, 0.799)$, which is consistent with the reference coordinates.

Table 1

Location of the primary recirculation centre of the lid-driven cavity case with $Re = 0.01$, $Wi = 0.5$ and $\beta = 0.5$.

Reference	x_p	y_p
Pan & Glowinski (2009)	0.469	0.798
Castillo & Codina (2015)	0.470	0.800
Souza et al. (2016)	0.467	0.801
Zhou et al. (2016)	0.468	0.798
Venkatesan & Ganesan (2017)	0.470	0.798
current work, CSF	0.467	0.799
current work, GLD-NSF	0.467	0.799

The agreement with literature results is corroborated by the velocity profiles along $x = 0.5$ and $y = 0.75$ shown in figure 7 which shows remarkable match of the current results with that from Venkatesan and Ganesan (2017).

To further assess the robustness of the proposed GLD-NSF formulation under more severe conditions, we simulated the lid-driven cavity with a Weissenberg number of $Wi = 1.0$, while maintaining the of $Re = 0.01$ and $\beta = 0.5$. It is well known that, as the Weissenberg number increases, approaches employing the standard Eulerian framework, such as the CSF formulation, can eventually be unable to deal with the numerical instabilities caused by the rapid growth of stress near the cavity corners causing the breakdown of simulations.

In order to ensure the mesh independence on results, we used the meshes shown in figure 9. Figure 10 presents

Table 2

Location of the primary (p) and secondary (s) recirculating regions of the lid-driven cavity. Parameters: $Re = 0.01$, $Wi = 1.0$ and $\beta = 0.5$.

Reference	x_p	y_p	$x_{s,1}$	$y_{s,1}$	$x_{s,2}$	$y_{s,2}$
Saramito (2014)	0.429	0.818	0.0364	0.0388	0.9637	0.0355
Pan et al. (2009)	0.439	0.816	-	-	-	-
Varchanis et al. (2019)	0.434	0.818	0.0375	0.0381	0.9637	0.0366
Chai et al. (2021)	0.438	0.817	0.0376	0.0387	0.9637	0.0371
current work, GLD-NSF	0.4290	0.8188	0.0376	0.0378	0.9652	0.0370

the temporal evolution of the kinetic energy, E_k , comparing the standard CSF approach with the proposed GLD-NSF method simulated with different mesh resolutions and time increments. Most notably, while the standard CSF method diverges and fails to yield a steady-state solution for the evaluated mesh configuration, the GLD-NSF formulation maintains numerical stability. The energy converges smoothly to a constant plateau, demonstrating the capability of the GLD-NSF approach. Furthermore, the overlapping kinetic energy profiles for different spatial resolutions ('Mesh 1' and 'Mesh 2') and time steps ($\Delta t = 1 \times 10^{-4}$ and 5×10^{-5}) confirms consistency of the proposed GLD-NSF formulation. This is also confirmed by figure 11 which shows the velocity profiles along $x = 0.50$ and $y = 0.75$ compared with the reference results from Fattal and Kupferman (2005).

The global flow topology for the cavity case with $Wi = 1.0$ is depicted via the streamlines in Figure 12. As the Weissenberg number increases to $Wi = 1.0$, the primary recirculation eye shifts upstream compared to the $Wi = 0.5$ case. Table 2 provides a quantitative comparison of the primary and secondary vortex coordinates with the data presented in literature. The GLD-NSF method predicts the primary vortex center at $(x_p, y_p) = (0.429, 0.819)$, which is in good agreement with the literature. The secondary vortices at the bottom corners are also well reproduced without any noticeable non-physical distortion in the flow field. We also note that references employed stabilization techniques such as the Log-Conformation Representation in Saramito (2014); Varchanis, Syrakos, Dimakopoulos and Tsamopoulos (2019); Pan et al. (2009) and Square-Root-Conformation in Chai, Ouyang and Wang (2021). This highlights how the GLD-NSF approach provides an alternative to these established strategies.

A more definite verification of the accuracy of the GLD-NSF formulation is given by the assessment of stress components profiles. Figures 13 and 14 present the stress components, T_{xx} , T_{xy} and T_{yy} , extracted along the cross-sections with $y = 0.75$ and $x = 0.5$, respectively. Even in regions characterized by steep stress gradients, the solutions do not exhibit spurious numerical oscillations. The perfectly overlapping profiles obtained using different spatial discretizations definitively confirm that the GLD-NSF framework not only achieves stability but does so while preserving rigorous mesh-independent accuracy.

6. Conclusions

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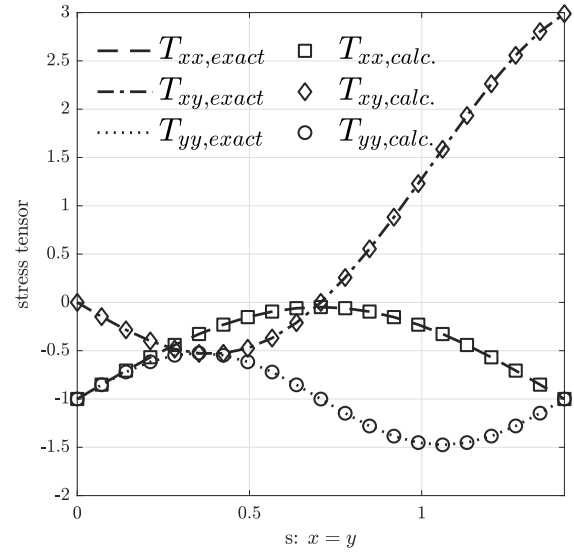


Figure 1: Stress tensor solution along $x = y$ obtained by the GLD-NSF method compared with the exact solution. $Wi = 0.25$, $\beta = 0.75$.

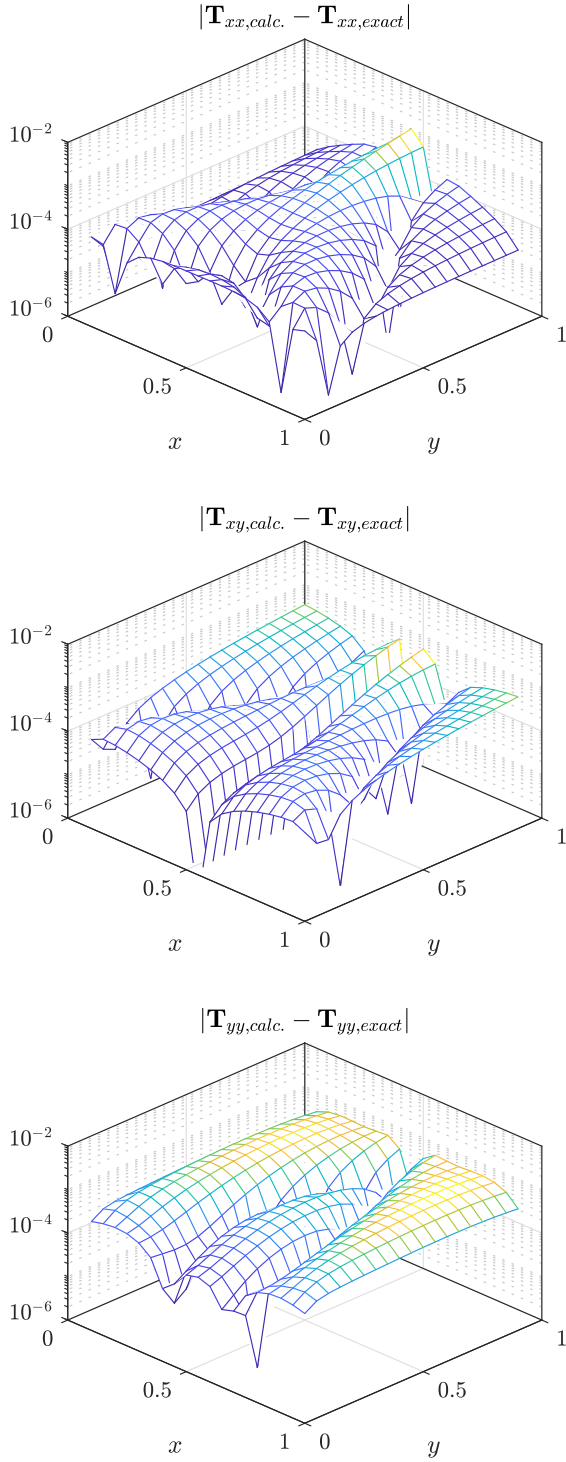


Figure 2: Contours of the absolute error of the stress tensor given by the GLD-NSF method relative the exact solution. $Wi = 0.25$, $\beta = 0.75$.

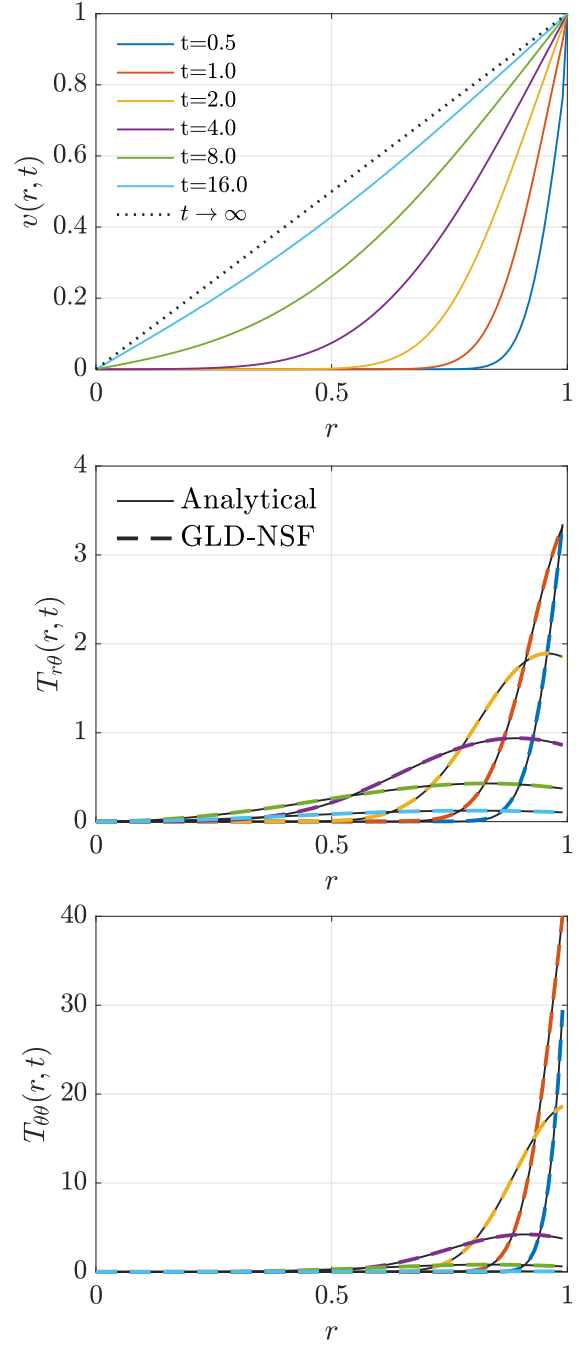


Figure 3: Comparison between analytical solution and numerical solution (GLD-NSF) for the start-up rotating cylinder. $Re = 100$, $Wi = 0.5$ and $\beta = 0.5$. $N = 600$, $dt = 0.001$.

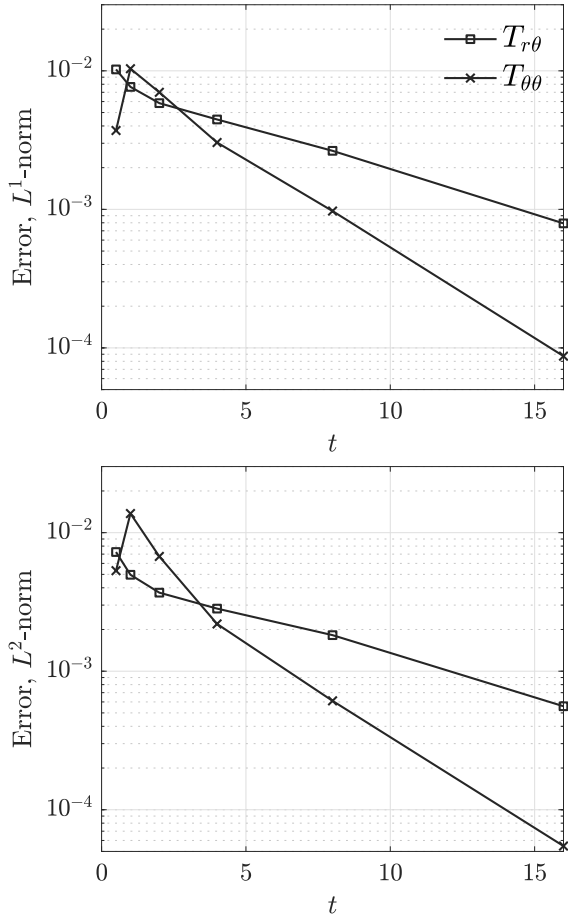


Figure 4: Error analysis of the numerical solution using GLD-NSF method for the start-up rotating cylinder. $Re = 100$, $Wi = 0.5$ and $\beta = 0.5$. $N = 600$, $dt = 0.001$.

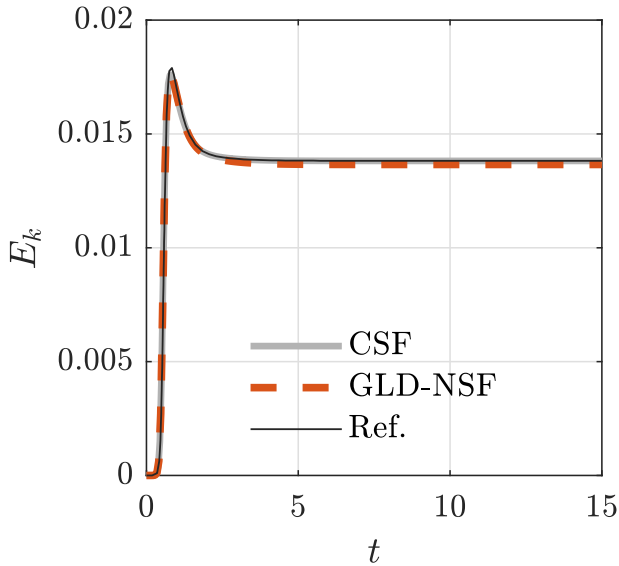


Figure 5: Time evolution of the elastic energy, E_k , of the current simulations for the lid-driven cavity compared with that shown by Venkatesan and Ganesan (2017) for the same case. Parameters: $Re = 0.01$, $Wi = 0.5$ and $\beta = 0.5$.

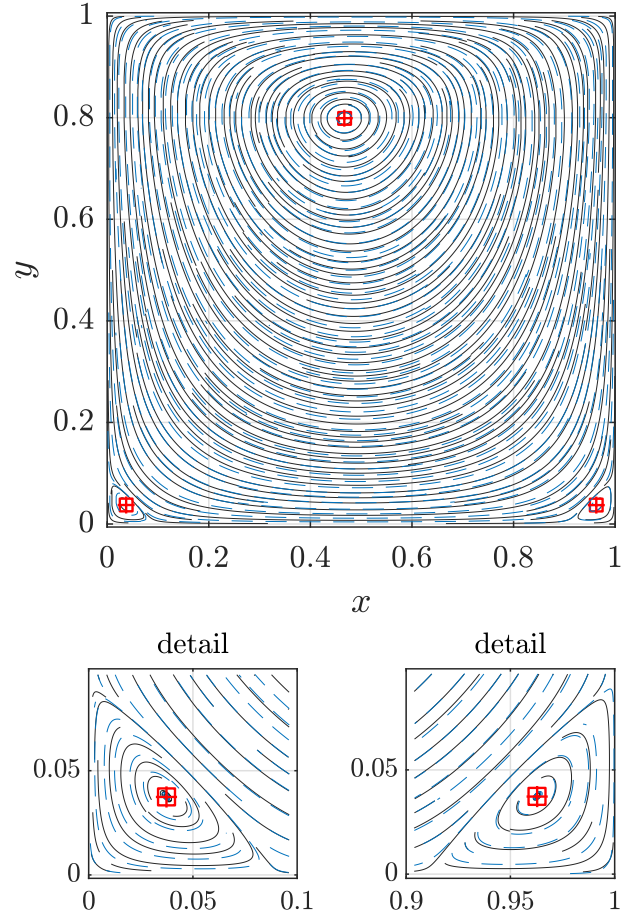


Figure 6: Streamlines of the lid-driven cavity obtained by methods CSF (solid) and the GLD-NSF (dashed) for $Wi = 0.5$. Markers shown in the plots (CSF: squares, GLD-NSF: cross) indicate the location of the primary and secondary recirculating regions also shown in detail. Parameters: $Re = 0.01$, $Wi = 0.5$ and $\beta = 0.5$.

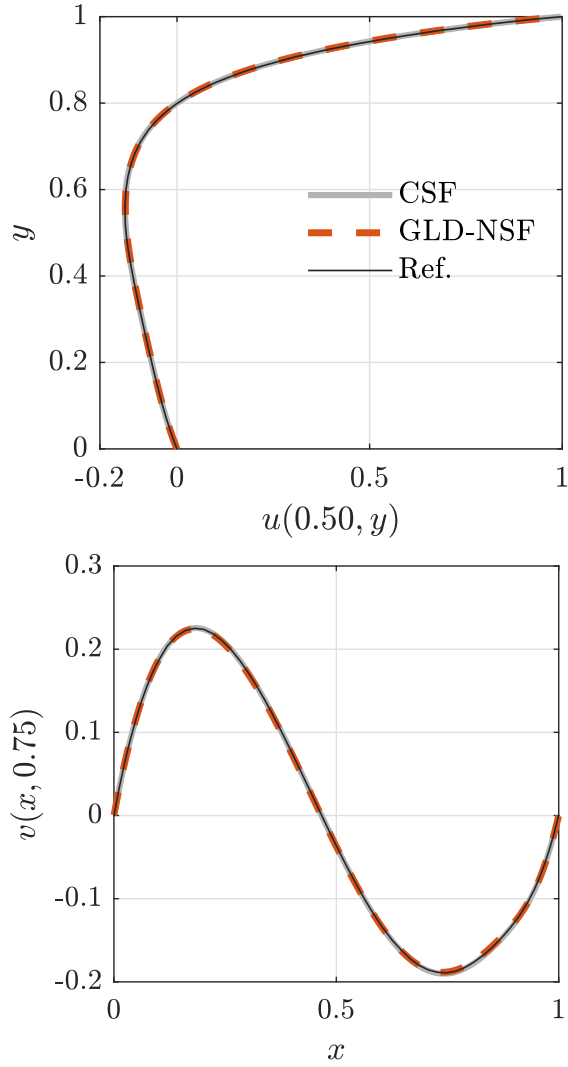


Figure 7: Velocity profiles at $x = 0.5$ and $y = 0.75$ of the current simulations for the lid-driven cavity compared with the results presented by Venkatesan and Ganesan (2017) for the same case. Parameters: $Re = 0.01$, $Wi = 0.5$ and $\beta = 0.5$.

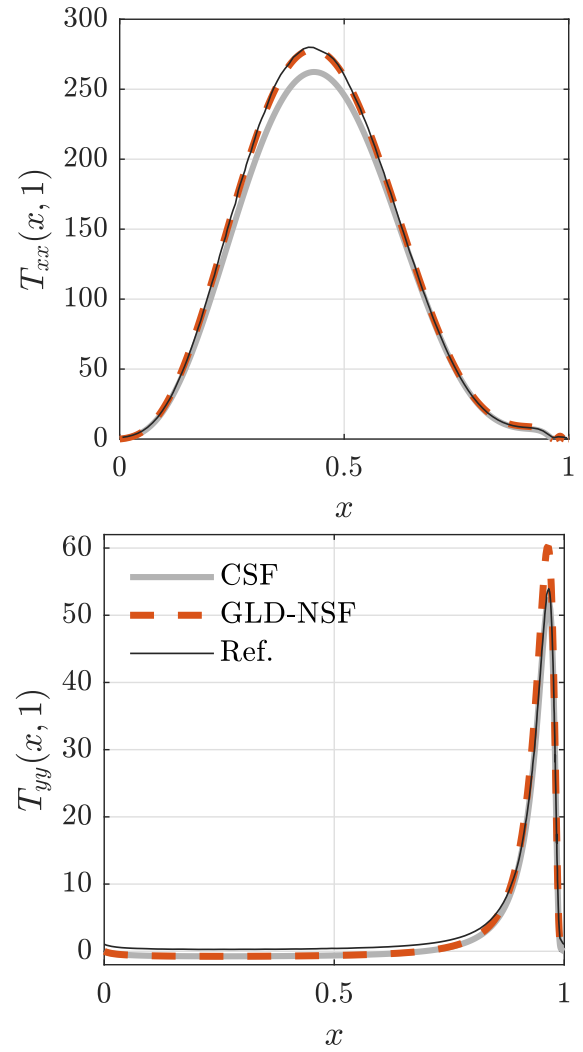


Figure 8: Stress tensor components, T_{xx} and T_{yy} , along the cavity top lid ($y = 1.0$) of the current simulations compared with that presented by Venkatesan and Ganesan (2017). Parameters: $Re = 0.01$, $Wi = 0.5$ and $\beta = 0.5$.

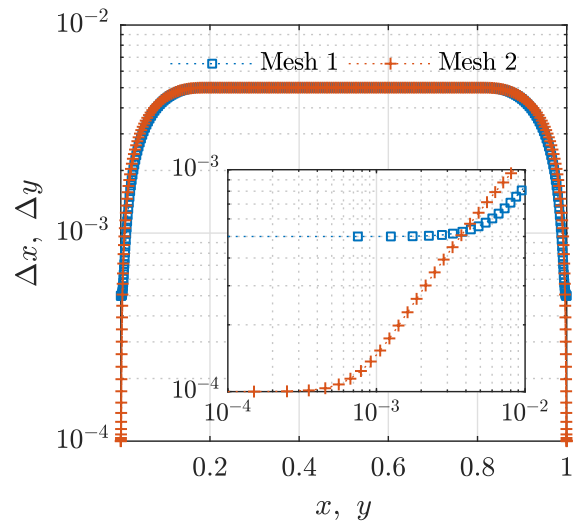


Figure 9: Meshes used in the current simulations.

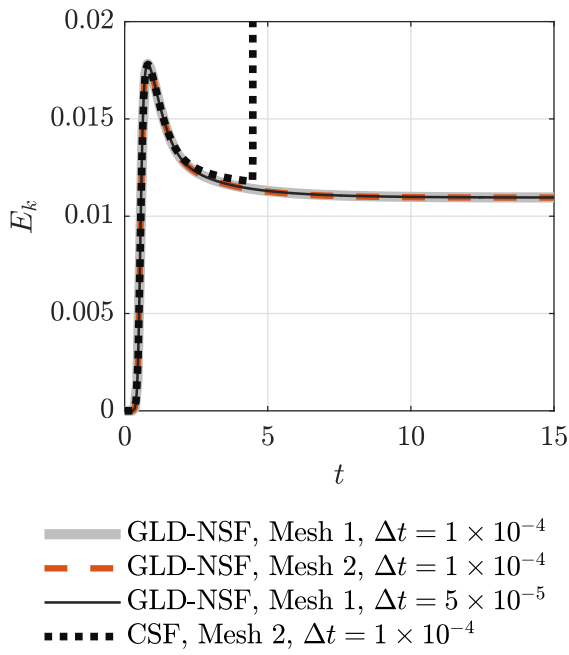


Figure 10: Time evolution of kinetic energy of lid-driven cavity obtained with CSF and GLD-NSF methods for different meshes and time increment. Parameters: $Re = 0.01$, $Wi = 1.0$ and $\beta = 0.5$.

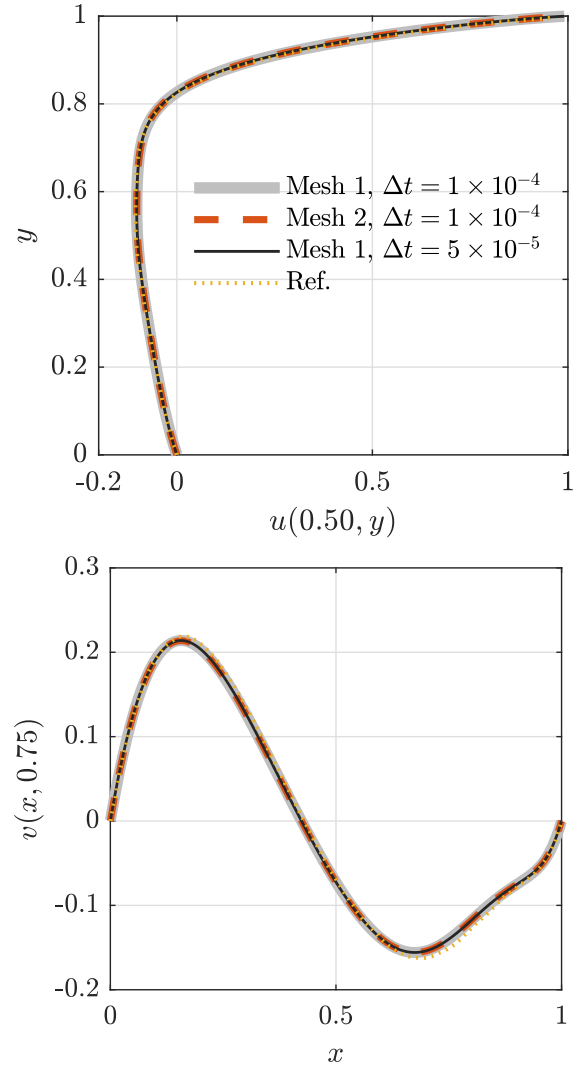


Figure 11: Velocity profiles at $x = 0.50$ and $y = 0.75$ of the current simulations for the lid-driven cavity compared with the results presented by Fattal and Kupferman (2005) for the same case. Parameters: $Re = 0.01$, $Wi = 1.0$ and $\beta = 0.5$.

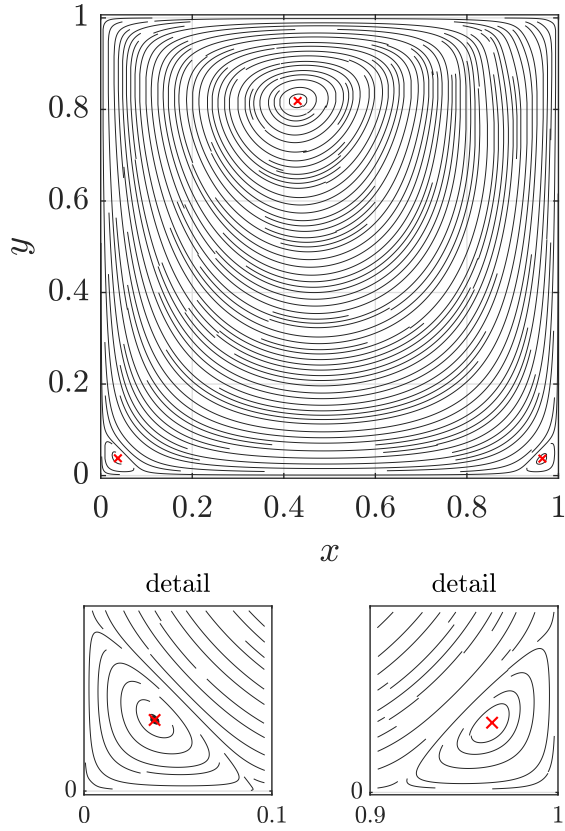


Figure 12: Streamlines of the lid-driven cavity obtained by GLD-NSF method. The red crosses indicate the centre of the primary and secondary recirculating regions (in detail). Parameters: $Re = 0.01$, $Wi = 1.0$ and $\beta = 0.5$.

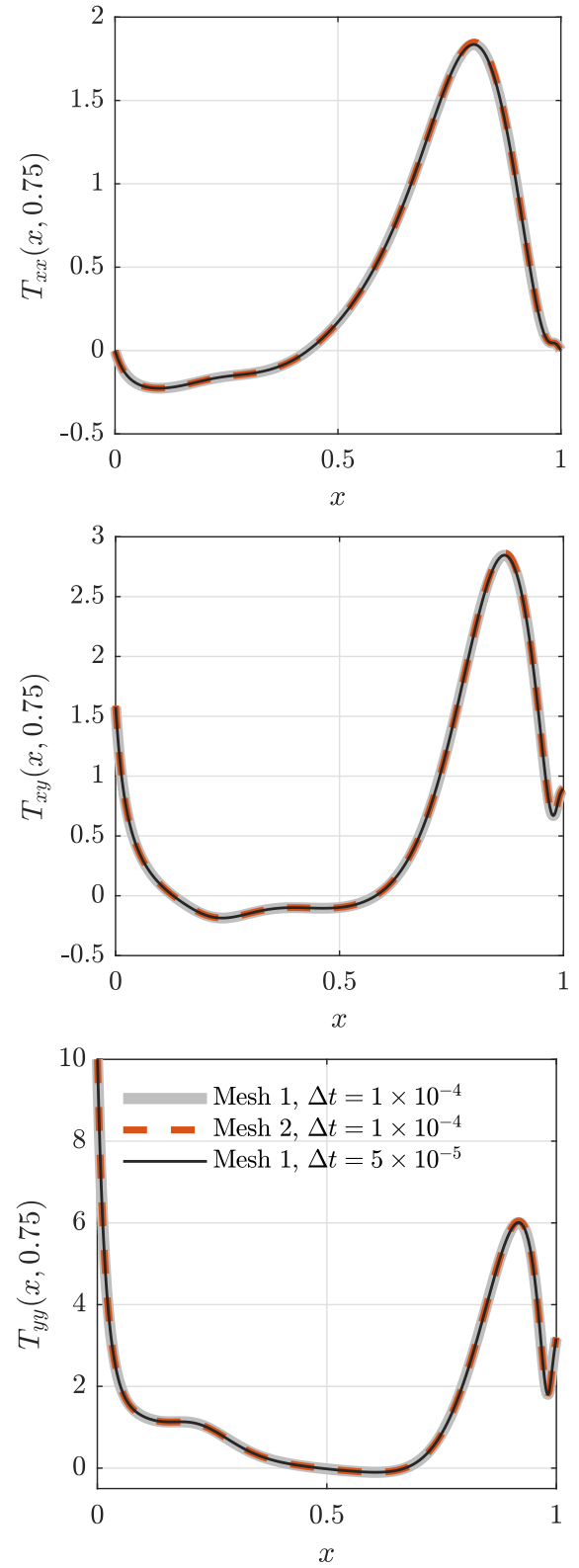


Figure 13: Stress tensor components along the $y = 0.75$ obtained from GLD-NSF simulations for different meshes. Parameters: $Re = 0.01$, $Wi = 1.0$ and $\beta = 0.5$.

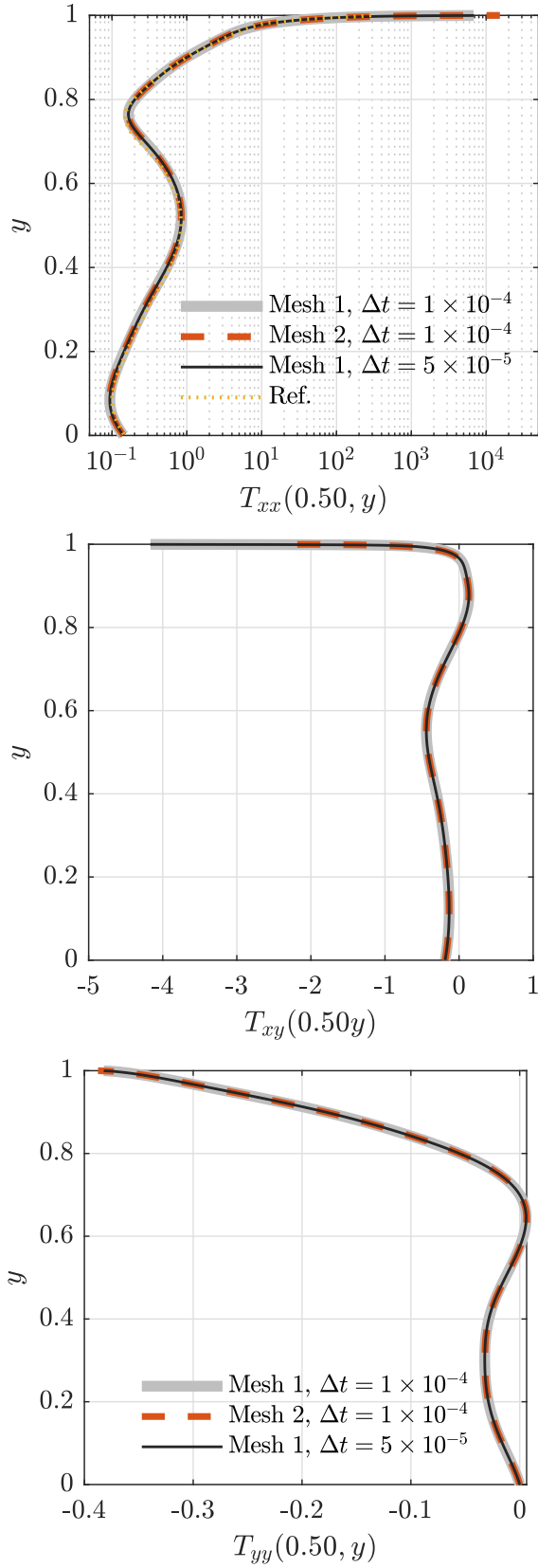


Figure 14: Stress tensor components along the $x = 0.5$ obtained from GLD-NSF simulations for different meshes. For T_{xx} , the curves are also compared with the reference result shown by Martins et al. (2015). Parameters: $Re = 0.01$, $Wi = 1.0$ and $\beta = 0.5$.